From Scores to Supported Claims: Measurement Audits of Document and Retrieval Evaluation

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Abstract— Evaluation scores are often used to compare systems, yet their interpretation depends on what an evaluator observes, how it scores the observation, and which comparison the evidence supports. We examine these dependencies through three measurement audits: a controlled staged document evaluator, the released FinRank evidence-ranking benchmark, and researcher-generated retrieval runs on two BEIR datasets. In the staged evaluator, 96 of 118 header changes were visible at the selected evidence surface; under a single-responsive-leaf restriction, flat and hierarchical scores retained the same ordering up to an approximate 0.375 rescaling despite different fixed-threshold detection rates. In FinRank, alternative tie policies changed absolute pairwise accuracy while preserving the direction of the hard-versus-random comparison. In the BEIR runs, a paired sign-flip analysis with Holm correction supported two of ten lexical-system comparisons on SciFact and one on NFCorpus. An exact-tie-aware point-estimate baseline already passed our duplicate and degradation controls. Together, these cases show how an audit can qualify or preserve a comparison by making its measurement contract explicit. The findings apply to the examined constructions, not to a universally validated audit method.

Index Terms— measurement validity, retrieval evaluation, reproducibility, document processing, benchmark auditing.

I. Introduction

A score is useful when readers know what it supports. A lower detection rate might indicate weaker ranking, or it might result from applying the same threshold to differently scaled scores. A changed benchmark score might alter a model-selection conclusion, or it might leave the comparison of interest intact. Likewise, the highest

mean on a retrieval leaderboard identifies a point-estimate leader but does not by itself establish that every apparent ordering is supported by the data.

These examples share a practical problem: a numerical result and the conclusion drawn from it need not describe the same measurement. We ask **which interpretations of an evaluation result remain supported when its observation surface, scoring rules and comparison procedure are made explicit?** We answer this question through three complementary audits, rather than proposing another general theory of validity.

Our first case uses a frozen staged document evaluator and controlled financial-table perturbations to separate observable changes from predicate detection. The second inspects FinRank [4] to distinguish tie-policy sensitivity in absolute scores from stability of a specific hard-versus-random comparison. The third audits our own retrieval runs on two BEIR datasets [5], asking how metric definitions, tie ordering and paired comparisons affect the interpretation of their scores. These are researcher-generated BEIR runs, not the published BEIR leaderboard; a later dense run is exploratory.

The contribution is an artifact-grounded account of how a reported result becomes a supported claim. In the examined constructions, we (1) explain a large detection-rate gap through a score rescaling under an explicit single-leaf restriction; (2) show that a benchmark comparison can retain its direction despite a material change in absolute scores; and (3) contrast point-estimate rankings with the support obtained from a specified family of paired statistical comparisons. The evidence also includes controls that limit each interpretation. Table 1 provides the resulting audit-to-interpretation map.

This work builds on prior studies of metric and construct validity [1]–[3]. AgentSuite’s component-oriented benchmark audit [11] and Chen’s study of oracle failures [12] address related validity and evaluation-system questions at broader Agent-evaluation scale. Our focus is narrower: the arithmetic, observation and comparison contracts governing particular document and retrieval results. A small development-only Agent inspection is retained in Appendix A as an illustration of the same observation/reference distinction, not as a fourth validated case.

II. Audit Questions and Procedure

For an evaluation object we ask: (i) whether a controlled change is visible at a specified evidence surface; (ii) whether the evaluation's predicate detects it; and (iii) whether it changes the downstream outcome under an explicit equivalence contract. These are different observations. A lack of visible change is not, by itself, evidence of harmful blindness.

Table 1. From an initial interpretation to the observation supported by each audit.

Evaluation case	A possible initial interpretation	Audit observation	Supported interpretation
Staged document evaluator	A lower detection rate necessarily indicates poorer ranking.	Only one leaf responds; scores are positively rescaled.	Account for score rescaling and report the active measurement surface.
FinRank	A tie-policy score shift means the hardness conclusion changes.	Scores change while the tested random-versus-hard gap retains its direction.	Separate score sensitivity from decision stability.
BEIR-based runs	Every higher mean score supports a strict ordering.	Some comparisons lack support under a paired, multiplicity-adjusted test.	Report the supported comparisons and uncertainty; non-significance is not equality.

These are interpretations of constructed or released evaluation objects, not observed changes in a real user's decision.

For scoring we separately vary declared conventions and inspect whether the reported absolute scores and the decision under examination change. Where systems are compared on shared queries, we use paired query-level resampling and report multiple-comparison sensitivity. We retain the original outputs and distinguish prospectively fixed choices from source-informed or post-hoc analyses.

The three applications differ in design and evidence. Case 1 uses synthetic perturbations on public tables. Case 2 inspects a released benchmark and its implementation. Case 3 uses independently generated retrieval runs over BEIR’s released data and relevance judgments; its original audit choices were not independently timestamped before the first run. All three support only the observations stated for their respective constructions.

The intervention, evidence surface, predicate and downstream outcome need to refer to compatible objects. For example, changing a stored program without recomputing the field that the predicate reads can produce a reproducible zero response without testing the predicate’s behavior on the changed program’s true result. We report such execution boundaries explicitly. For pairwise ranking, the comparison and scoring conventions are part of the measurement contract, and conclusions about the direction of a comparison are separate from conclusions about its numerical magnitude.

III. Staged Document Evaluator

The staged evaluator has parsing and question-answering checkpoint trees containing twelve scored leaves. Its seed-42 primary panel contains 95 public TAT-QA-derived source tables [7] with 118 controlled header changes across six families. The observed surface consists of mapping and regression artifacts. In the reported comparison, the flat protocol uses the field-hit-rate drop, while the hierarchical expert protocol uses a recursively weighted parsing-root drop; detection is based on those drops under the reported fixed operating points. A change is visible when that surface records a difference, while protocol detection and downstream harm remain separate judgments. The surface includes only the selected artifacts, not every internal model state or every possible user outcome.

Of the 118 injected changes, 96 were visible. None of 13 OCR-like changes appeared on the selected surface. This is an observation boundary, not proof of 13 harmful misses: one existing semantic judge classified the cases as referentially equivalent. The QA tree was frozen and the field-hit-rate leaf was the only responsive leaf in this header-change construction. Under this restriction, the unrounded hierarchical contrastive score was a positive rescaling of the flat score, with fitted slope approximately 0.375 and identical observed ranks. For these observed scores, corresponding thresholds approximately satisfy $\theta_H = 0.375\ \theta_F$; applying fixed

operating points without corresponding rescaling can yield different detection rates despite the same ordering. This is an observed structural equivalence in the present construction, not a general result for multiple responsive leaves.

Table 2. Seed-42 detection and stage-instrumentation controls.

Evaluation and unit	Flat / full response	Hierarchical / direct or reduced response	Interpretation
Header detection, 95 tables at the reported operating points	Flat 75/95	Hierarchical expert 20/95	Raw rates depend on threshold and score scale; ranks match in this single-leaf construction.
Header detection, 118 injections at the reported operating points	Flat 85/118	Hierarchical expert 13/118	Injection and table denominators are different units.
Stage localization, 216 controlled injections	Full vector macro-F1 and top-1 1.000	Three direct leaves: 1.000; removing them: top-1 0.333	Stage alignment is provided by explicit instrumentation.

The relationship can be read directly as a change of scale. On the matched-threshold reanalysis grid, a flat drop threshold of 1.0 corresponds to a hierarchical root-drop threshold of 0.375; for example, the recorded evaluation fold detects 21 cases under each of these corresponding thresholds. This grid example illustrates matched decisions; it is not the operating-point configuration used for the 75/95 and 20/95 rows in Table 2.

The clean-only calibration has a separate limitation. In the frozen score construction, an unmodified clean record is compared with a repeated deterministic computation on the same record, so its contrastive score is zero by construction. All 2,382 usable clean records have zero scores for both protocols. Consequently, every positive threshold has an observed clean false-positive rate of zero, and this panel does not identify a unique deployment threshold. More clean

records evaluated under the same zero-by-construction definition would not by themselves resolve this ambiguity.

Across three seeds under the flat protocol, the descriptive detection table-run ratio was 219/282 (75/95, 76/97, and 68/90), using 119 unique source-table identifiers. A post-hoc UID-cluster bootstrap, which retains repeated table appearances within each source, gives a 95% interval of [0.7117, 0.8364]. Equal weighting of source UIDs gives 0.7535 [0.6835, 0.8207]. The per-seed denominators (95, 97 and 90) are not the count of one shared panel, and three seeds are insufficient to characterize a population of future seeds. The semantic assessment above relies on one judge rather than an independent human reference.

A separate stage-aligned instrument introduced controlled mapping, retrieval and numeric changes. The three stages are mapping, retrieval and numeric processing. Each of 72 source clusters contributes one controlled injection per stage. For the binary leaf-response vector, a leave-one-cluster-out nearest-centroid classifier attained macro-F1 and top-1 accuracy of 1.000 over 216 injections. The same separation followed from the three direct stage-targeted leaves: parsing, retrieval and reconciliation. Removing these leaves reduced top-1 accuracy to 0.333 with a 0.667 tie rate under the frozen tie-as-incorrect rule; the remaining leaves retained partial information but did not separate retrieval from numeric changes. Thus the perfect controlled separation is attributable to the explicit instrumentation, rather than to hierarchy alone. A nested FinQA probe [8] changes program fields without recomputing downstream result fields and is interpreted only as a measurement-object boundary.

The stage study uses leave-one-cluster-out evaluation, with one injection from each of the three contracts per cluster; its positive result is for the frozen controlled set. The FinQA boundary study contains 89 rows in 30 report clusters. It records 89 visible changes, no detected change and no changed outcome field; because the downstream results were not regenerated, those zeros do not estimate the detector’ s true failure rate on changed answers.

IV. FinRank: Score Sensitivity and Decision Stability

FinRank is a public evidence-ranking benchmark over financial filings [4]. We inspect its released implementation and outputs, including exact ties and source identity. The tie comparisons and thresholds were fixed before pair-level outcomes. A later canonical-text identity correction is explicitly post-hoc; the source code made the inspected identity behavior visible before this analysis.

Under tie-as-positive scoring, an exact score tie counts for the gold passage; under strict-win scoring it does not. Table 3 shows that this change moves random-negative accuracy by 1.084 percentage points for TF-IDF and 1.029 for BM25, above the fixed 1.0-point materiality threshold. Yet the random-minus-hard gap remains positive for both systems: the evaluated qualitative conclusion survives even though the absolute scores are sensitive to the tie contract.

Table 3. FinRank pairwise accuracy and random-minus-hard gap under two tie policies. Values are percentages; gaps and movements are percentage points.

Model	Tie policy	Hard accuracy	Random accuracy	Random minus hard
TF-IDF	Tie-as-positive	77.756	91.935	14.179
TF-IDF	Strict win	77.030	90.850	13.821
BM25	Tie-as-positive	74.780	87.789	13.009
BM25	Strict win	74.094	86.760	12.666

Movements and gaps are calculated before display rounding; subtracting the rounded cells need not reproduce the reported values exactly. This result concerns the stated comparison, rather than every possible use of the scores.

The corrected, source-informed identity analysis reproduces two known properties: 442 hard-negative occurrences across 305 records share canonical text identity with another record’ s gold passage, and the inspected metadata filter loses gold evidence in 33 records. In the pooled corpus, text is deduplicated by first occurrence, whose metadata is retained; filtering on that metadata can

consequently exclude a passage that is gold for a later record. These counts are post-hoc reproductions, not newly discovered or independently preregistered benchmark defects.

In-record and pairwise metrics reproduce exactly in the audited environment. Some global and metadata-filtered metrics still differ from released results; without the original score matrices or complete environment, the cause remains unresolved. Numerical reproduction and ranking interpretation are distinct claims.

V. Public Retrieval Evaluation

We independently constructed five lexical retrieval configurations on BEIR SciFact and NFCorpus [5]: TF-IDF and BM25 with (k1, b) values (0.9, 0.4), (1.2, 0.75), (0.5, 0.3) and (1.5, 0.6). The evaluation uses the released test queries, corpora and relevance judgments: 300 judged queries on SciFact and 323 on NFCorpus. These are two subsets of one benchmark family, not two independently established benchmarks. The original five-system analysis is retrospective and covers no dense or generative retriever.

During development our own first implementation binarized graded relevance and used a different MAP denominator. Comparing it with the trec_eval contract corrected both definitions. Under graded nDCG@10, TF-IDF moved from second to third in the NFCorpus example ranking; its position change is specific to the constructed systems and judgments, not a correction to an official leaderboard. The reference implementation trec_eval 10.0-rc3 was subsequently run on the fixed rankings. Across 18,690 per-query nDCG, MAP and recall values, discrepancies were within 1e-4, with maximum approximately 5e-5 due to printed precision. The comparison verifies scores for given rankings, not the policy that breaks ties.

These are two separate corrections: graded relevance changed nDCG, while dividing average precision at the cutoff by the full number of relevant documents changed MAP@10 on NFCorpus from approximately 0.2225 to 0.1193 for the example. The latter does not explain the nDCG ranking change. The cross-check used trec_eval itself, not the pytrec_eval Python binding; it does not demonstrate agreement with a separately executed pytrec_eval installation.

We compared the five systems using 2,000 paired bootstrap resamples and a two-sided paired sign-flip test with 2,000 Monte Carlo draws per pair and Holm

correction across ten pairs. The uncorrected bootstrap intervals excluded zero for three pairs on SciFact and four on NFCorpus; the corrected sign-flip analysis supported two and one, respectively. These are separate inferential procedures, so the change in counts cannot be ascribed solely to Holm correction. A failure to exclude zero is not proof that two systems are equal.

Changing cutoff from k=10 to k=100 also affected system ordering. Separately, for one BM25 configuration on NFCorpus at k=10, reversing document-ID tie order changed the ordered top ten for 36.5% of queries and changed top-ten membership for 25.7%. Mean nDCG@10 changed from 0.30453945 to 0.30449093, about 0.00005; the aggregate rounds to the same four-decimal display. On SciFact those tie-change rates were zero. The finding illustrates a per-query effect hidden by aggregation within these conditions, not a general failure rate.

We also compared naive sorting, a point-estimate rule that treats exactly equal scores as ties, and the paired audit. The naive view incorrectly asserts an order for an intentionally duplicated system; the simple equality rule and audit both avoid that assertion and both identify an intentionally degraded control. Thus this control does not show the audit beats the simple rule. The audit reports which comparisons are supported under its statistical contract, at the cost of additional metric sweeps and resampling; it is not a demonstrated real-user decision improvement.

Table 4. Evidence about retrieval comparisons.

Object	Observation	Limit of the interpretation
Five lexical systems, SciFact / NFCorpus	Holm-adjusted paired sign-flip tests support 2/10 and 1/10 pairs.	The ten pairs are within each dataset; no conclusion of equality follows from a non-significant comparison.
NFCorpus tie-order perturbation, one BM25 system	36.5% of queries change ordered top-10; 25.7% change its membership; mean nDCG@10 changes by about 0.00005.	These are query-list, query-set and aggregate-score quantities, respectively; this is not an official benchmark-wide failure rate.
Exact duplicate and degraded controls	Exact-tie point estimates and the paired audit both pass both controls.	These controls do not establish an advantage over the simpler baseline.

Post-hoc dense exploration and six-system sensitivity analysis. To widen model coverage, we subsequently ran the public BGE-small-en-v1.5 encoder [9], using the same data, qrels and nDCG@10 contract. We normalized embeddings, used the model's retrieval query prefix, capped inputs at 512 tokens and compared with the fixed BM25(1.2, 0.75) run. Dense versus BM25 mean scores were 0.7127 versus 0.6622 on SciFact and 0.3438 versus 0.3062 on NFCorpus. The paired bootstrap differences were 0.0505 [0.0137, 0.0891] and 0.0376 [0.0154, 0.0608], respectively; Table 5 displays the same mean differences to five decimal places. Each interval is uncorrected and the exploration was chosen after the original analyses; neither belongs to the original ten-pair Holm family or establishes a general model ranking. Independent execution of trec_eval agreed for all 1,246 dense and BM25 per-query scores to its four-decimal print precision. The encoder improvement is a retrieval result, not evidence that the audit improved a user decision or that the lexical tie conclusions transfer to dense models.

To examine multiplicity sensitivity in the combined family, we subsequently assessed six retrieval runs, including BGE and five lexical configurations, using the saved query-level outcomes. Under graded nDCG@10, we examined all 15

unordered pairs per dataset. Two-sided paired sign-flip tests used 200,000 Monte Carlo draws per pair and Holm correction within each dataset's 15-pair family. The procedures assume within-query exchangeability under a sharp null or suitable symmetry of paired differences, and query independence for inference. Neither assumption is established by the numerical recheck.

Table 5. Exploratory BGE versus BM25(1.2, 0.75) graded nDCG@10 comparison in the six-system family, with 200,000 sign-flip draws per pair.

Dataset	Queries	Mean graded nDCG@10 difference, BGE minus BM25	Unadjusted two-sided sign-flip p	Holm-adjusted p, 15 pairs
SciFact	300	+0.05049	0.006410	0.066000
NFCorpus	323	+0.03759	0.001295	0.015540

Holm step-down adjustment enforces monotonicity over the ordered p-values within each family. At $\alpha = 0.05$, this selected SciFact comparison does not meet the family-adjusted threshold, whereas the NFCorpus comparison does. The result on SciFact does not prove equality. Analyses were selected after the earlier results were known; this is neither independent confirmation nor evidence of a person changing a decision. Holm adjustment was performed separately for each dataset and metric condition, not across datasets or conditions. The recheck reduces Monte Carlo variability but does not verify the statistical assumptions; its methods and results have not received comprehensive independent statistical review. Saved dense data contain ranked top-100 document IDs without original dense scores, so they do not establish dense-score tie frequencies.

VI. Discussion and Limitations

The cases connect evaluation objects to supported claims. A visible mutation is not automatically detected; a non-response is not automatically harmful; metric sensitivity is not automatically a decision reversal. Explicit stage instrumentation produces a controlled positive result, while its direct-leaf control limits what may be attributed to hierarchy. FinRank's decision remained stable under the examined tie policies, even as absolute scores moved. The BEIR development

correction shows how an evaluation contract can change an example ranking, while careful point-estimate handling already solves the duplicated control.

Scope is restricted. Header changes are synthetic on one source corpus. FinRank covers lexical baselines and has a residual reproduction gap. BEIR covers two subsets of one benchmark and constructed lexical systems, plus one post-hoc dense encoder; its design is retrospective. The supplementary Agent inspection comprises four source-exposed runs in two settings and does not establish agent generalization; we did not test human decisions, generative retrieval or deployment benefit. Confidence intervals and multiplicity correction do not establish equivalence when a comparison is inconclusive. Existing construct-validity frameworks already cover broad audit principles [2], [3]; our contribution is the bounded application and the observed distinctions, not a general theory.

VII. Conclusion

Three measurement audits illustrate how evidence surfaces, score conventions and reproducibility delimit evaluation claims. In the tested constructions, explicit checkpoints aided controlled stage separation; a FinRank comparison retained direction across tie policies; and a BEIR-based example showed that graded relevance and uncertainty can alter interpretation. These findings motivate reporting the measurement object, scoring contract, uncertainty and reproduction scope together. General effectiveness and deployment value remain open.

Appendix A. Development-only Agent inspection

The following source-exposed checks are included as a supplementary illustration, not a fourth independent audit case. BASE and PROSE denote two previously generated migration candidates, not two independent repositories. For each selected SciFact query, A and B are the two presented candidate passages.

Post-hoc Agent inspection (development only). To make the evidence boundary concrete, an isolated OpenCode–Qwen agent inspected two existing generated candidates from a real NtripBrowser dependency migration [10] and made two two-document SciFact choices. Both migration candidates passed the same seven historical local tests, but a fixed direct-import/call contract classified BASE as invalid and PROSE as valid. The Agent read the files and gave the same verdicts as

this inexpensive AST check; this is not evidence of an agentic-method advantage. On SciFact queries 13 and 36, it selected passages unjudged by the available qrels, matching the BM25 choices. Unjudged is not an independently established negative. Both queries were deliberately selected for this incomplete-judgment condition, and all sources were previously exposed.

Table A1. Development-only Agent inspections.

Inspection object	Final answer	Additional observation or reference	Interpretation boundary
Migration BASE / PROSE	INVALID / VALID	Historical tests 7/7 for both; direct-call AST INVALID / VALID; completed file reads	The AST and Agent apply the same public narrow contract; neither provides a complete semantic ground truth, and historical tests were not rerun here.
SciFact query 13 / 36	A / B	Both selections unjudged in qrels and match BM25; query and both candidate reads recorded	No conclusion that the selected passages are irrelevant or that the Agent used the returned content.

The four Agent runs used eight local model HTTP requests and reported 31,054 tokens including cached input. Their traces and fixed reference are retained for offline reanalysis; the exact deployed weight hash was not independently verified. These are two demonstration settings (one repository and two selected queries), not four independent external benchmark replications. The AST check recognises direct literal import and calls only; it does not resolve aliases or verify runtime consumption. AgentSuite [11] and the oracle-integrity study [12] already investigate benchmark components and evaluator failures at larger scale; the above examples

are a limited illustration of this paper’ s score–observation–reference distinction, not a new general Agent-auditing method.

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